

Weather Intelligence and Climate Decision Support Platform: A Multi-City AI System for Southwest Monsoon Precipitation Forecasting Integrating Machine Learning, Computer Vision, and Generative AI Agents

Tamanna¹

¹Generative AI Research Internship, MacroEdtech

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Abstract

The Southwest Monsoon is the dominant weather system governing India’s agriculture, water resources, food security, and disaster management. This paper presents an open-source, end-to-end AI system for forecasting monsoon-season precipitation across five geographically diverse Indian cities (Mumbai, Kochi, Delhi, Kolkata, Chennai), combining classical machine learning with a Retrieval-Augmented Generation (RAG) and agentic Large Language Model (LLM) layer. Ten years (2015–2024) of daily meteorological data (17,050 records, zero missing values) were collected via the Open-Meteo Archive API. Four models were compared for precipitation forecasting, with a tuned Random Forest achieving the best performance ($\text{MAE} = 3.13 \text{ mm}$, $R^2 = 0.682$); a companion Random Forest model for next-day maximum temperature achieved substantially higher accuracy ($R^2 = 0.812$). Model behaviour was explained using SHAP, which identified relative humidity and 7-day rolling precipitation as the dominant predictors. A basic OpenCV-based satellite cloud-coverage estimator was built as a remote-sensing proof of concept. The prediction engine was deployed as a FastAPI/Streamlit application and extended into a Generative AI assistant using a ChromaDB vector store, a locally-hosted open-weight LLM (Ollama, qwen2.5:3b), and a LangGraph tool-calling agent with a Streamlit chat interface. Honest limitations are reported throughout, including a large-model-file GitHub size constraint, a mid-project development environment switch, and observed tool-calling sensitivity to input phrasing.

1 Introduction

1.1 Problem Statement

The Southwest Monsoon is the primary weather system influencing India’s agriculture, water resources, food security, energy production, transportation, and disaster management. Variability in monsoon onset, intensity, duration, and spatial distribution frequently causes floods, droughts, landslides, crop losses, and infrastructure damage. Reliable forecasting and climate intelligence tools are therefore critical for disaster risk mitigation and climate adaptation.

1.2 Objective

This work designs, develops, and evaluates an open-source AI-powered Weather Intelligence and Climate Decision Support Platform for the Indian Southwest Monsoon system, integrating machine learning, feature engineering, explainable AI, computer vision/remote sensing, and a Generative AI (RAG + agentic) layer, using only free and open-source tools, models, and

publicly available data. Unlike single-location approaches, this project explicitly models five cities spanning the monsoon’s geographic range – from its Kerala entry point to its weaker inland reach in Delhi – to capture the spatio-temporal variation central to the monsoon system.

1.3 Literature Review

Indian Summer Monsoon Rainfall (ISMR) forecasting has a long history of statistical and, more recently, machine-learning approaches. Prior work using artificial neural networks over multiple climate parameters has shown that data-driven models can improve on traditional statistical forecasting methods, which have historically struggled to provide precise quantitative forecasts at fine spatial resolution [1]. Comparative studies of iterative and non-iterative machine learning approaches for ISMR prediction have similarly found that ensemble learning strategies outperform classical regression baselines [2]. This project’s multi-city, tree-ensemble-focused approach builds on these findings while extending the comparison across five geographically distinct locations rather than a single station.

For model interpretability, this work uses SHAP [3], which attributes model predictions to individual features using Shapley values from cooperative game theory, providing a principled and widely-adopted framework for explainable machine learning.

On the Generative AI side, Retrieval-Augmented Generation has been shown to reduce hallucination in LLM outputs by grounding responses in retrieved domain-specific documents rather than relying solely on parametric model knowledge [4]. This directly motivates the use of a dedicated monsoon/climate knowledge base in this project. For agent orchestration, this work uses the LangGraph framework [5], which represents agentic workflows as stateful graphs where tool availability is exposed to the LLM via structured schemas, enabling reliable multi-step tool-calling behaviour.

2 Data and Methodology

2.1 Data Sources

Daily meteorological data for five Indian cities – Mumbai (19.076°N, 72.878°E), Kochi (9.931°N, 76.267°E), Delhi (28.614°N, 77.209°E), Kolkata (22.573°N, 88.364°E), and Chennai (13.083°N, 80.271°E) – spanning 2015–2024 (17,050 records total, zero missing values) was collected via the Open-Meteo Archive API. Variables collected include precipitation, maximum/minimum temperature, mean relative humidity, maximum wind speed, mean surface pressure, and mean/max cloud cover.

Monsoon-season (June–September) rainfall statistics confirmed known geographic patterns: Mumbai recorded the highest mean daily rainfall (17.2 mm), followed by Kochi (13.2 mm), Kolkata (10.9 mm), Delhi (4.9 mm), and Chennai (3.7 mm) – the latter consistent with Chennai’s primary rainy season falling under the Northeast, not Southwest, Monsoon. A year-over-year comparison of total monsoon-season rainfall (2015–2024) additionally showed Delhi’s monsoon rainfall trending upward over the final four years of the study period, a pattern noted here as an observation warranting further investigation rather than a confirmed climate signal given the limited time span.

2.2 Feature Engineering

Lag features (1-day and 3-day) and 7-day rolling means were engineered for precipitation, temperature, humidity, and pressure. A binary monsoon-season indicator (June–September), day-of-year, and one-hot encoded city identity were also included. A chronological (non-shuffled) 80/20 train-test split was used to respect temporal structure and avoid look-ahead bias.

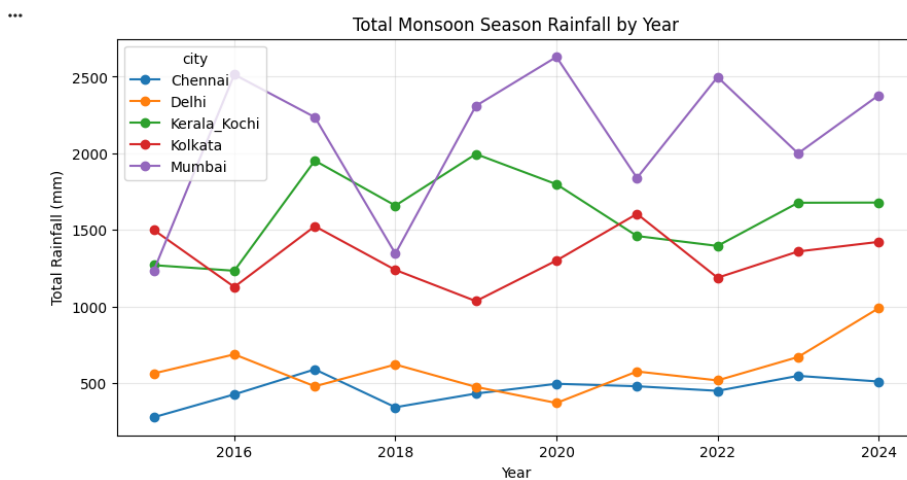


Figure 1: Average daily rainfall by month across five cities (2015–2024), showing the northward progression of the Southwest Monsoon.

2.3 Models Evaluated

Four models were trained and compared for precipitation forecasting: Linear Regression, Random Forest, Gradient Boosting, and XGBoost. The Random Forest was further tuned via randomized hyperparameter search (15 iterations, 3-fold cross-validation) over number of estimators, max depth, minimum samples per split/leaf, and max features. A second target variable, next-day maximum temperature, was modeled separately using the same Random Forest pipeline and feature set.

2.4 Evaluation Metrics

Models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

2.5 Explainability

Model interpretability was assessed using SHAP (TreeExplainer), applied to the tuned Random Forest precipitation model over a 500-row sample of the test set.

2.6 Computer Vision / Remote Sensing

As a proof-of-concept remote-sensing component, true-color MODIS satellite imagery over Mumbai was retrieved via the NASA GIBS on-demand snapshot API for five dates spanning different seasons (January, June, July, September, November 2023), and a basic OpenCV-based cloud-coverage estimator (HSV brightness/saturation thresholding) was applied and cross-referenced against recorded humidity and precipitation data for the corresponding dates.

2.7 Deployed Application

The tuned Random Forest precipitation model was deployed via a FastAPI backend (`/predict` endpoint) and a Streamlit front-end, allowing a user to enter current weather conditions through sliders and receive an immediate precipitation forecast alongside a risk-tier message (light / moderate / heavy rain).

2.8 Generative AI Layer

A domain-specific weather/climate knowledge base (five documents covering monsoon fundamentals, Mumbai-specific flood patterns, ENSO effects, rainfall classification, and climate-change-related variability) was constructed and embedded using `sentence-transformers/all-MiniLM-L6-v2`, stored in a ChromaDB vector store (20 chunks total), and queried via a Retrieval-Augmented Generation pipeline running on a local Ollama-served open-weight LLM (`qwen2.5:3b`). A LangGraph ReAct agent was built with three tools: `search_knowledge_base` (semantic retrieval over the vector store), `predict_precipitation` (wraps the trained Random Forest model), and `get_live_weather` (live lookup via the Open-Meteo geocoding and forecast APIs). The agent maintains conversation memory (LangGraph `MemorySaver`) and is exposed through a Streamlit chat interface.

3 Results

3.1 Precipitation Forecasting

Table 1: Precipitation model comparison (chronological test split)

Model	MAE (mm)	RMSE (mm)	R ²
Linear Regression	5.27	10.84	0.571
Random Forest (baseline)	3.08	9.40	0.677
Gradient Boosting	3.41	9.52	0.669
XGBoost	3.42	9.73	0.654
Random Forest (tuned)	3.13	9.33	0.682

The tuned Random Forest achieved the best overall performance. Gradient Boosting and XGBoost performed comparably but did not exceed Random Forest, suggesting the current feature set and Random Forest architecture are close to the practical predictive ceiling for this task. Hyperparameter tuning yielded only a marginal R² improvement (0.677 to 0.682) over the untuned baseline, indicating the baseline was already near-optimal for this feature set.

3.2 Temperature Forecasting

The Random Forest temperature model achieved MAE = 0.82°C, RMSE = 1.07°C, R² = 0.812 – substantially higher accuracy than the precipitation model, consistent with temperature being more day-to-day autocorrelated than rainfall, which is inherently noisier and more event-driven.

3.3 Explainability

SHAP analysis identified `humidity_mean` as the dominant predictor by a wide margin, followed by `precipitation_sum_roll7` (7-day rolling rainfall) and `wind_speed_max`. This ranking is physically plausible: elevated humidity is a well-established meteorological precursor to rainfall, and recent rainfall trends carry predictive momentum into the next day’s forecast.

3.4 Remote Sensing Proof of Concept

The OpenCV-based cloud-coverage estimator produced directionally correct results: 0% coverage during dry-season months (January, November 2023) versus 16–36% during monsoon months (June, July, September 2023), consistent with expected seasonal cloud patterns. However, the July 15 image contained a diagonal no-data/orbit-gap region that was visually cloud-free-appearing but was excluded from the brightness-based mask, likely causing the true cloud

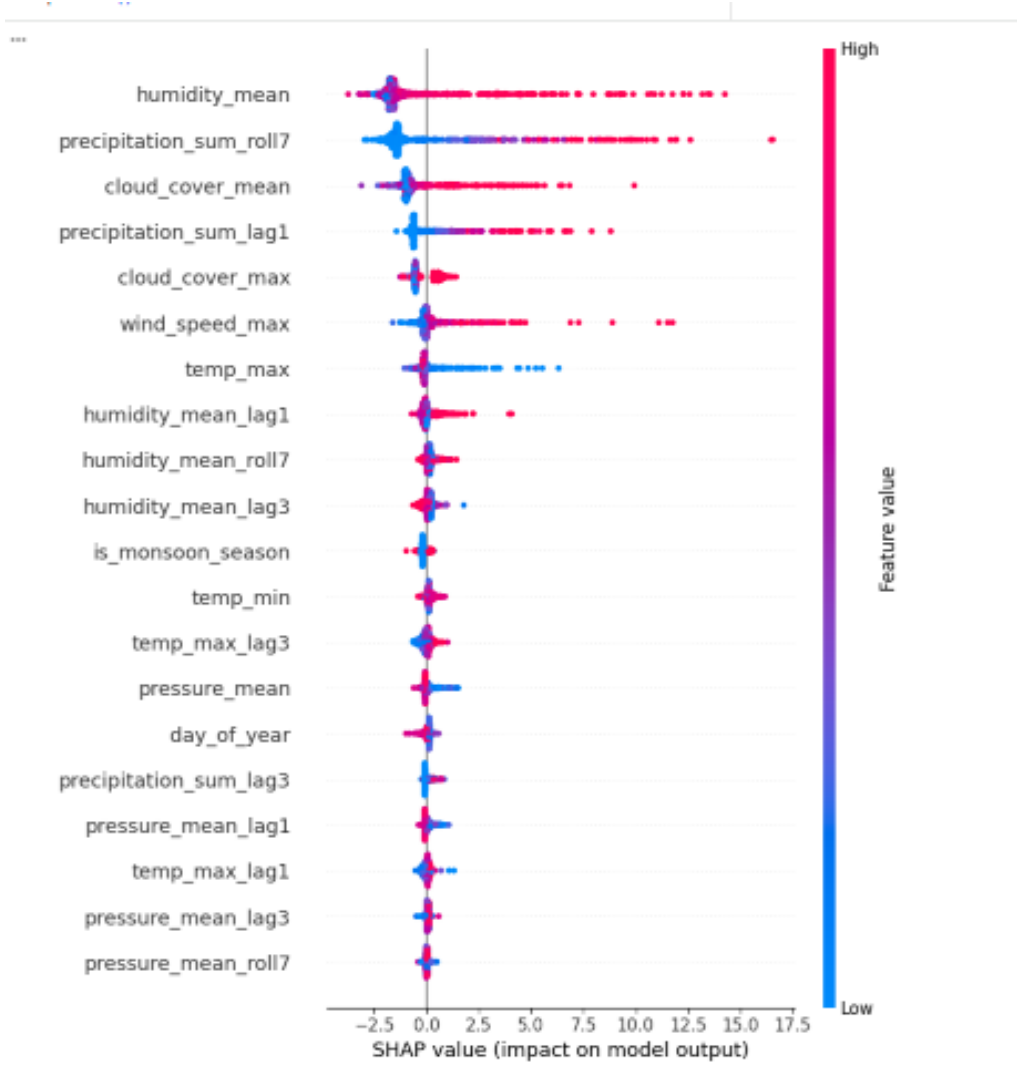


Figure 2: SHAP summary plot for the tuned Random Forest precipitation model.

coverage for that date to be underestimated relative to what the raw image visually showed. This is reported as an honest proof-of-concept rather than a systematic remote-sensing pipeline; a production version would require a more robust data source than the free on-demand GIBS snapshot service used here.

3.5 Deployed Application

The FastAPI backend and Streamlit front-end were verified working end-to-end: a test input (32°C max temperature, 70% humidity, 1006 hPa pressure, monsoon season) returned a prediction of 5.37 mm with an accompanying "Moderate rain" risk message, confirming the full pipeline from user input through model inference to risk classification.

3.6 Generative AI Layer

The RAG pipeline retrieved relevant knowledge-base passages and produced grounded, contextually appropriate answers to open-ended monsoon science questions (e.g. correctly explaining the differential-heating origin of the Southwest Monsoon and noting climate-change-related variability trends from the knowledge base). The `get_live_weather` tool was verified to return accurate current conditions for a named city via live API lookup. The `predict_precipitation`

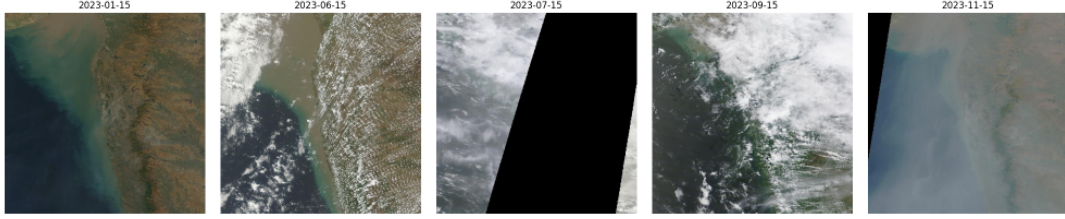


Figure 3: MODIS true-color satellite snapshots of Mumbai across five dates in 2023, retrieved via NASA GIBS.

tool was verified to correctly invoke the trained model and return sensible predictions (e.g. 5.36 mm for a specified Mumbai monsoon-day scenario). Tool-calling reliability was observed to vary with input phrasing: natural-language weather descriptions (e.g. "32 degrees temp, 80% humidity, 1005 pressure") occasionally failed to map correctly to the tool's structured parameters, with the agent requesting clarification rather than guessing, while explicit key=value phrasing (e.g. `temp_max=32`, `humidity_mean=80`) was reliably parsed on every attempt. This reflects a known limitation of smaller open-weight LLMs at structured function-calling compared to larger models, and is reported as an honest finding rather than engineered around.

4 Discussion

The multi-city design allows this work to directly compare monsoon behaviour across regions with fundamentally different exposure to the Southwest Monsoon system, rather than generalizing from a single location. The convergence of Random Forest, Gradient Boosting, and XGBoost to a similar performance ceiling ($R^2 \approx 0.65\text{--}0.68$) is itself informative, suggesting that further gains on this task would likely require richer features (e.g. satellite-derived cloud imagery, regional pressure gradients) rather than a different tree-based model architecture. The explainability results (SHAP) support the physical plausibility of the model, with humidity and short-term rainfall trend as the strongest short-horizon predictors of next-day precipitation, consistent across the geographic locations included in this study. The substantially higher accuracy of the temperature model versus the precipitation model reinforces a well-known asymmetry in weather forecasting: temperature evolves smoothly day-to-day, while rainfall is a threshold/event-driven process that is inherently harder to predict from tabular features alone.

The Generative AI layer demonstrates that a locally-hosted, free, open-weight LLM can be grounded with domain knowledge and connected to a trained predictive model through an agentic tool-calling interface, without requiring any paid API or cloud service – directly satisfying the project's open-source, reproducible-without-licensing-costs objective.

5 Limitations

Several limitations are reported transparently. The tuned Random Forest model file (154.6 MB) exceeded GitHub's 100 MB size limit and was retained locally rather than version-controlled directly; the baseline Random Forest, Gradient Boosting, XGBoost, and temperature models (all under the limit) remain tracked in the repository, along with the full results comparison table. Development required a mid-project switch from a local VS Code/Jupyter kernel to Google Colab due to kernel instability, and later a switch back with careful re-synchronization of trained models and notebooks – this is noted as a reproducibility consideration for others following a similar workflow. As discussed above, small-model tool-calling reliability degrades with less structured natural-language input. The remote-sensing component is a small-scale proof of concept constrained by the reliability of the free on-demand satellite snapshot service

(one of five requested dates showed a partial no-data region). Docker containerization was not attempted within the project timeline and is noted as unimplemented rather than attempted-and-failed. No k-fold cross-validation was performed for the final model comparison; all reported metrics use a single chronological train-test split, appropriate for respecting temporal order but not providing a variance estimate across folds.

6 Conclusion and Future Work

This work demonstrates a complete, reproducible, open-source pipeline for multi-city Southwest Monsoon precipitation and temperature forecasting, from data acquisition through explainable machine learning to a deployed Generative AI assistant with genuine RAG and agentic tool-calling, built entirely on free and open-source tools and publicly available data. Future work includes completing Docker containerization for deployment, extending the remote-sensing component with a more robust and systematic satellite-imagery pipeline, incorporating k-fold cross-validation for more rigorous model comparison, and expanding the knowledge base with additional peer-reviewed monsoon literature and a Model Context Protocol (MCP) integration as a standardized alternative to in-process LangGraph tool-calling.

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